

1. Administrative & Study Identification

Full Study Title: Potential Eligibility and Estimated Preventable Cardiovascular Disease Events From Inclisiran Treatment in the United States
Short Title/Acronym: US Inclisiran Eligibility & Preventable ASCVD Events (NHANES Study)
Protocol Number: CKJX839D1US03
NCT Number: NCT07214857
Sponsor Name: Novartis Pharmaceuticals
Investigational Product: Inclisiran
Phase of Development: N/A (Observational / Retrospective Database Study)
Medical Monitor: N/A (Observational Database Study)

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To estimate the number of United States (US) adults with and without prior atherosclerotic cardiovascular disease (ASCVD) and diabetes who may be eligible for inclisiran based on US National Health and Nutrition Examination Survey (NHANES) data and inclisiran eligibility criteria from ongoing clinical trials. **Secondary Objectives:** To estimate the number of preventable ASCVD events over 10 years based on expected risk reductions from the degree of low-density lipoprotein cholesterol (LDL-C) lowering expected from inclisiran.

2.2 Background & Rationale Many patients do not reach recommended low-density lipoprotein cholesterol (LDL-C) levels on statins alone. Inclisiran is a small interfering RNA providing a ~50% further reduction in LDL-C. This study aims to address the gap in understanding the population-level impact of inclisiran by quantifying the number of eligible patients and estimating the potential cardiovascular events that could be prevented with its use in the US over a 10-year horizon.

3. Study Design & Methodology

3.1 Design Framework Study Type: Observational (Cross-sectional study using NHANES 2011-2020 survey data) **Control Type:** N/A (Population-based statistical modeling) **Blinding:** Open-label / N/A **Randomization:** N/A

3.2 Planned Enrollment & Duration Target \$N\$: 1,385 (NHANES participant sample, weighted to represent the US adult

population) **Number of Sites:** N/A (US National Database) **Estimated Study Start:** 29-Mar-2024 **Estimated Primary Completion:** 09-Oct-2024

4. Subject Selection (Eligibility Criteria)

- 4.1 Inclusion Criteria** 1. Aged 18 years of age or older. 2. Elevated LDL-C ≥ 70 mg/dL (≥ 55 mg/dL if very high-risk for ASCVD events) despite baseline statin therapy. 3. Patients falling into one of three populations: Secondary Prevention (known ASCVD), High-Risk Primary Prevention ($\geq 20\%$ 10-year ASCVD risk or 7.5% to $<20\%$ with ≥ 2 risk-enhancing factors), or Diabetes Population.
- 4.2 Exclusion Criteria** 1. Patients under 18 years of age. 2. Missing LDL-C, laboratory, or required covariate data in the NHANES dataset to calculate the 10-year ASCVD risk. 3. Patients whose LDL-C is adequately controlled on current statin therapy per current guidelines.
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5. Intervention & Treatment Plan

- 5.1 Investigational Regimen Dosage/Frequency:** N/A for an observational study (Modeled based on standard inclisiran subcutaneous administration twice yearly after the initial doses). **Route of Administration:** Subcutaneous (Modeled) **Duration of Treatment:** Modeled over a 10-year ASCVD risk prevention horizon.
- 5.2 Concomitant & Prohibited Therapy Permitted:** Baseline statin therapy and other background lipid-lowering therapies. **Prohibited:** N/A (Retrospective database analysis).
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6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures		
---	---	Screening	N/A	Application of inclusion/exclusion criteria to NHANES 2011-2020 datasets
		Baseline	N/A	Calculation of baseline 10-year ASCVD risk using ASCVD Pooled Cohort Equation and SMART2 risk scores
		Interim	N/A	N/A
		End of Study	N/A	Final extrapolation of preventable ASCVD events over 10 years

(Note: Because this is a retrospective database study, there are no actual patient visits or clinical schedules.)

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Estimated number of US adults eligible for inclisiran. **Measurement Method:** Application of inclisiran trial eligibility criteria to the NHANES 2011-2020 weighted population.

7.2 Secondary & Exploratory Endpoints Estimated number of preventable ASCVD events over 10 years. **Measurement Method:** Calculated using the ASCVD Pooled Cohort Equation and SMART2 risk score "without inclisiran", and applying inclisiran trial data and Cholesterol Treatment Trialists' data "with inclisiran" to estimate reductions in ASCVD events per mmol/L of LDL-C reduction.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: N/A (This is a non-interventional, observational database study) **Serious Adverse Events (SAEs):** N/A **Data Monitoring Committee (DMC):** N/A

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Cross-sectional analysis of US adults ≥ 18 years from the NHANES 2011-2020 database. **Sample Size Justification:** The $N=1,385$ unweighted sample provides sufficient statistical power when applying NHANES survey weights to accurately represent approximately 20 million eligible US adults. **Handling of Missing Data:** Standard complex survey weighting and imputation methodology consistent with NHANES analytical guidelines.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study was conducted in accordance with guidelines for observational research and secondary data analysis. **Monitoring Plan:** N/A (Database extraction and statistical modeling study). **Audit Trail:** Documented statistical programming scripts and data extraction logs for generating population estimates.

11. Study Identifiers & Governance

Primary ID: CKJX839D1US03 **Registry Number:** NCT07214857 **Sponsor/Lead Organization:** Novartis Pharmaceuticals **Study Title:** Potential Eligibility and Estimated Preventable Cardiovascular Disease Events From Inclisiran Treatment in the United States **Official Regulatory Title:** Potential Eligibility and Estimated

12. Clinical Framework (ASCVD/Lipid Specific)

Primary Condition: Cardiovascular Disease / Atherosclerotic Cardiovascular Disease (ASCVD) **Diagnosis Threshold:** Elevated LDL-C ≥ 70 mg/dL (or ≥ 55 mg/dL for very high risk) despite statin therapy **Medical Methodology:** Cardiovascular risk estimation utilizing the ASCVD Pooled Cohort Equation and SMART2 risk score **Optimization Target:** $\sim 50\%$ further reduction in LDL-C levels via small interfering RNA therapy modeling

13. Study Procedures & Methodology

Management Pathway: Retrospective analysis of US national health surveys (NHANES 2011-2020) **Education Protocol (HCP):** N/A **Education Protocol (Patient):** N/A **Quality Audit Mechanism:** Peer-reviewed methodology for statistical extraction, epidemiological weighting, and event-reduction modeling

14. Observation & Follow-Up Schedule

Total Duration: 10-year modeled risk horizon **Onsite Visit Cadence:** N/A (Retrospective survey analysis) **Remote Monitoring Frequency:** N/A **Checklist Review Interval:** N/A

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]				Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]				